

**SIMATS ENGINEERING
ASSESSMENT TOOL 3**

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA1710

COURSE OUTCOME COVERED

***CO3:** Analyze knowledge representation and reasoning using propositional logic, FOL, inference, resolution, chaining methods, knowledge representation to perform logical reasoning for drawing valid conclusions and decision making. (BL4)*

Assessment Tool Weightage: 20%

QUESTIONS: 5

Time Duration: 1 Hour
Total Marks:30

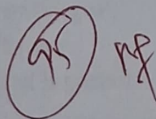
Annexure A

Resolution Algorithm Implementation

Code of Conduct:

*I V. RUPASHREE (192319012) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student: *V. Rupa*



Student 28: RUPASHREE V | Register Number: 192319012

Question 1 - Machine Auto Shutdown

Knowledge Base

1. If a machine overheats, then the machine shuts down automatically.
2. The machine overheated.

Goal

Prove that **the machine shuts down automatically.** using the Resolution Algorithm.

Tasks

1. Represent the statements in propositional logic.
2. Convert all statements into CNF. Negate the goal.
3. Apply the Resolution Algorithm step by step.
4. Show the Empty Clause (Box) if the goal is proved.

Question 2 - Account Verification

Knowledge Base

1. If a user uploads a valid ID, then the account is verified.
2. The user uploaded a valid ID.

Goal

Prove that **the account is verified.** using the Resolution Algorithm.

Tasks

1. Represent the statements in propositional logic.
2. Convert all statements into CNF. Negate the goal.
3. Apply the Resolution Algorithm step by step.
4. Show the Empty Clause (Box) if the goal is proved.

Question 3 - Premium Subscription Access

Knowledge Base

1. If a customer pays the subscription fee, then the customer gets premium access.
2. Anitha paid the subscription fee.

Goal

Prove that **Anitha gets premium access.** using the Resolution Algorithm.

Tasks

1. Represent the statements in propositional logic.
2. Convert all statements into CNF. Negate the goal.
3. Apply the Resolution Algorithm step by step.
4. Show the Empty Clause (Box) if the goal is proved.

Question 4 - Smart Device Connectivity

Knowledge Base

1. If a device is connected to WiFi, then the device can access the internet.
2. The smart TV is connected to WiFi.

Goal

Prove that **the smart TV can access the internet.** using the Resolution Algorithm.

Tasks

1. Represent the statements in propositional logic.
2. Convert all statements into CNF. Negate the goal.
3. Apply the Resolution Algorithm step by step.
4. Show the Empty Clause (Box) if the goal is proved.

Question 5 - Order Shipping Confirmation

Knowledge Base

1. If a payment is verified, then the order is confirmed.
2. If the order is confirmed, then the item is shipped.
3. The payment was verified.

Goal

Prove that **the item is shipped**. using the Resolution Algorithm.

Tasks

1. Represent the knowledge base in propositional logic.
2. Convert all implications into CNF.
3. Negate the goal.
4. Apply the Resolution Algorithm to derive the Empty Clause (Box).
5. Explain each resolution step and write the final conclusion.

Numerical Problems

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Marks
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem	15
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method	19
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process	24
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations	19
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation	16

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Machine Auto Shutdown:-

Knowledge base:-

1. If a machine overheats, then the machine shuts down automatically.
2. The machine overload.

Goal: Prove that the machine shuts down automatically.

1. Propositional logic:-

Let:

P = Machine overheats

Q = Machine shuts down automatically.

Statements:

$$P \rightarrow Q$$

$$P$$

2. Convert to CNF:-

$$P \rightarrow Q \Rightarrow \neg P \vee Q$$

So the clauses are:

$$C_1 = \neg P \vee Q$$

$$C_2 = P$$

3. Negate the goal

$$\text{Goal} = Q$$

$$\text{Negate goal} = \neg Q$$

$$C_3 = \neg Q$$

4. Resolution:-

Resolve $C1 \& C2$

$(\neg P \vee Q), P \rightarrow Q$

New clause:

Q
Now resolve:

$Q, \neg Q \rightarrow \square$

5) Conclusion:

Since the empty clause is obtained, the goal is proved.

Therefore, the machine shuts down automatically.

2) Account Verification:-

Knowledge base:

1. If a user uploads a valid ID, then the account is verified.

2. The user uploaded a valid ID.

Goal: Prove the account is verified.

1. Propositional Logic:

Let:

P = user uploads a valid ID

Q = Account is verified

Statements:

$P \rightarrow Q$

P

2. CNF:-

$$P \rightarrow Q = \neg P \vee Q$$

$$\text{Clause: } c_1 : \neg P \vee Q$$

$$c_2 : P$$

3. Negate the Goal:

$$\text{Goal} = Q$$

$$\text{Negated goal} = \neg Q$$

$$c_3 = \neg Q$$

4. Resolution:

$$c_1 : \neg P \vee Q$$

$$c_2 : P$$

Resolving:

$$(\neg P \vee Q), P \rightarrow Q$$

Then:

$$c_4 : Q$$

$$c_3 : \neg Q$$

$$Q, \neg Q \rightarrow \square$$

5. Conclusion:-

Empty clause is obtained.

Therefore, the answer is verified.

3. Premium Subscription Access :-

Knowledge Base:

1. If a customer pays the subscription fee, then the customer gets premium access.

Goal: Prove that Anirudh gets premium access.

1. Propositional Logic:-

Let:

P : Anirudh pays the subscription fee

Q : Anirudh gets premium access.

Statements:

$$P \rightarrow Q$$

P

2. CNF

$$P \rightarrow Q = \neg P \vee Q$$

Claims:

$$= \neg P \vee Q$$

$$C_1 = \neg P \vee Q$$

$$C_2 = P$$

3) Negate the goal:-

$$\text{Goal} = Q$$

$$\text{Negated goal} = \neg Q$$

$$C_3 = \neg Q$$

4) Resolution :-

$$C_1 : \neg P \vee Q$$

$$C_2 : P$$

$$\therefore Q$$

5. Conclusion:-

Since the empty clause $\square$ is obtained,

$\therefore$ Anima gets Premium Access.

4. Smart Device Connectivity:-

Knowledge Base:

1. If a device is connected to wifi, then the device can access the internet.

2. The smart TV is connected to wifi.

Goal: Prove that the smart TV can access the internet.

1. Propositional Logic:-

Let:

P = Smart TV is connected to wifi

Q = Smart TV can access the internet.

Statements:

$P \rightarrow Q$
 P

2. CNF

$$P \rightarrow Q = \neg P \vee Q$$

Clauses:-

$$C_1 = \neg P \vee Q$$

$$C_2 = P$$

3. Negate the goal.

$$\text{Goal} = Q$$

$$\text{Negated goal} = \neg Q$$

$$C_3 = \neg Q$$

4. Resolution:-

$$(\neg P \vee Q), P \rightarrow Q$$

$$Q, \neg Q \rightarrow \square$$

5. Conclusion:-

The empty clause $\square$ is obtained.

$\therefore$ The smart TV can access the internet.

5. Vehicle Registration Process:-

Knowledge base:-

1. If a car passes the emission test, then the car is certified roadworthy.

2. If a car is certified roadworthy, then the car can be registered.

3. The car passed the emission test.

Goal: ~~Prove~~ that the car can be registered.

1. Propositional Logic:-

Let:

P = car passes the emission test

Q = car is certified roadworthy

R = car can be registered.

Statements:

$$P \rightarrow Q$$

$$Q \rightarrow R$$

$$P$$

convert to CNF :-

$$P \rightarrow Q = \neg P \vee Q$$

$$Q \rightarrow R = \neg Q \vee R$$

Clauses

$$C_1 : \neg P \vee Q$$

$$C_2 : \neg Q \vee R$$

$$C_3 : P$$

3) Negate the goal

$$\text{Goal} = R$$

$$\text{Negated goal} = \neg R$$

$$C_4 : \neg R$$

4) Resolution steps

$$C_1 : \neg P \vee Q$$

$$C_3 : P$$

Resolve P & $\neg P$:

$$C_2 : \neg Q \vee R$$

$$C_5 : R$$

Resolve R & $\neg R$:-

$\therefore$ The car can be registered.